

~~Q1.~~ Given the dataset below, check which fits better – a Gaussian distribution, or a Gamma distribution. Use the Method of Moments to estimate the necessary parameters.

D1	D2	D3	D4	D5	D6	D7	D8	D9	D10
23	14	7	3	9	39	10	1	1	4
D11	D12	D13	D14	D15	D16	D17	D18	D19	D20
5	5	14	7	4	3	9	3	15	3

~~Q2.~~ You have a coin with bias 0.5, which you don't know and are trying to estimate by Maximum Likelihood. If you do 10 tosses, how likely is it, that your estimate from these tosses has an error at most 0.1?

~~Q3.~~ You are given 3 observations, which are 2D vectors of the form $(X \ Y)$. The observations are $[(1.7 \ -1.6), (2.3 \ -1.9), (-0.4 \ 0.8)]$. Under the assumption that they follow 2D Gaussian, estimate their mean vector and covariance matrix.

~~Q4.~~ For any random variable X with $X \sim N(m, s^2)$, $P(m < X < m+s) = 0.34$ and $P(m+s < X < m+2s) = 0.135$ (roughly). Suppose you have 100 observations from a Gaussian distribution whose variance parameter is known to be 25, and the sample mean of the observations comes to 10. Specify a range, such that you are at least 95% confident that the true mean of the Gaussian must lie within that range.

[Hint: consider the distribution of the statistic corresponding to the sample mean of 100 observations]

~~Q5.~~ You are given a 6-faced dice which you initially believe to be fully fair. You carry out 2 rounds of 10 throws each, where the outcomes (frequency of faces) are as below:

	#1	#2	#3	#4	#5	#6
Round 1	3	2	0	0	2	3
Round 2	2	1	1	0	4	2

Using suitable probability distributions, show how your belief about the fairness of the dice gets updated after each round.

~~Q6.~~ There is a Gaussian distribution $N(u, 9)$. Consider a prior distribution $N(0, 25)$ on ' u '. You receive data sequentially from the Gaussian distribution $[5.1, 4.3 \ 7.7 \ 5.4 \ 2.1 \ 4.8]$. Show how the posterior and MAP estimation of ' u ' evolve as this data arrives one by one? Derive the necessary expressions.

~~Q7.~~ Let's say, I have divided the samples into two sets: A ($x_1, \dots, x_n$) and B ($x_{n+1}, \dots, x_N$). I calculate the MLEs of the mean separately from A (n samples) and B (N-n samples), and then take the mean of these two estimates as my final estimate. Is the distribution of this estimate same or different from the usual case, where all N samples are used together to estimate 'u'?

~~Q8.~~ Consider the following model: $Y_i \sim \text{Ber}(p)$, $Z_i \sim U(0,1)$, $X_i \sim N(W_k Z_i, s)$ where $k = Y_i$. We have 3 observations of X and Z: [$X_1=3.6$ $Z_1=0.2$, $X_2=-1.2$ $Z_2=0.5$, $X_3=4.2$ $Z_3=0.8$].

Consider the following 2 choices of parameters: P1: [$W_1=-5$, $W_2=5$, $s=9$, $p=0.5$], P2: [$W_1=0$, $W_2=4$, $s=25$, $p=0.8$]

Which of these two sets of parameters fits the observations better? We do not observe Y, so we must marginalize over all possible values of Y.

~~Q9.~~ Consider the following model: $Y_i \sim \text{Ber}(p)$, $Z_i \sim N(0,1)$, $X_i \sim N(W_k Z_i, s)$ where $k = Y_i$. We have 3 observations of X and Y: [$X_1=3.6$ $Y_1=1$, $X_2=-1.2$ $Y_2=0$, $X_3=4.2$ $Y_3=1$]. Consider the following 2 choices of parameters:

P1: [$W_0=-5$, $W_1=5$, $s=9$, $p=0.5$], P2: [$W_0=0$, $W_1=4$, $s=25$, $p=0.8$]

Which of these two sets of parameters fits the observations best? Note that we do not observe Z, so we must marginalize over all possible values of Z by integration.

~~Q10.~~ Consider you have training data of the form (X, Y). You want to fit the relation $Y=WX+e$ on them, for which you have put the prior $N(0, 9)$ on W. Now you receive 5 observations: [(5, 11), (2, 4), (-4, -7), (7, 15), (-3, -5)]. Assuming $e \sim N(0,1)$, what is now your posterior on W? Derive the necessary expression (considering 1D data).

~~Q11.~~ Consider an autoregressive process where $X_t \sim N(aX_{t-1}, s^2)$. From N observations $X_1, X_2, \dots, X_N$, how can I make maximum-likelihood estimate of a and s^2 ? How can I make a Bayesian estimate of a, assuming s^2 is known?

~~Q12.~~ We have N data-points of the form $\{x_i, y_i\}$ where x_i, y_i are both real numbers. I want to fit a model $Y = WX + E$ where W is clearly a real number and E is a Gaussian noise $N(0, s^2)$. I have a prior distribution $p(W) = N(0, 1)$ on the parameter W. Estimate the posterior distribution $p(W|D)$. Extend the above analysis to the case where X, W are D-dimensional real numbers.